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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Introduction to Graph Algorithms (course)

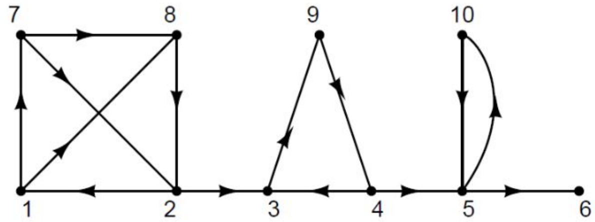
If already registered, click to check your payment status

Course outline

Week 8 Assignment 8

Your last recorded submission was on 2025-09-17, 09:56 Due date: 2025-09-17, 23:59 IST. IST

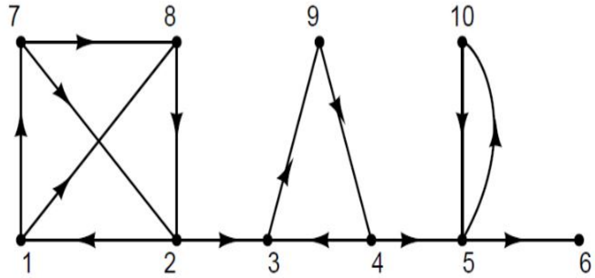
1) Let G be the ___ for the given graph. ***2 points***



The vertices in the largest strongly connected components are {___}

- ○ {1,2,7,8}
- ○ {1,2,3,7,8}
- ○ {1,2,5,8}
- ● {1,2,5,7,8}

2) Let G be the ___ for the given graph.



The number of strongly connected components are __

4

2 points

Week 4 Bellman Ford and Dijkstra Algorithm ()

Week 5 All Pair Shortest Path ()

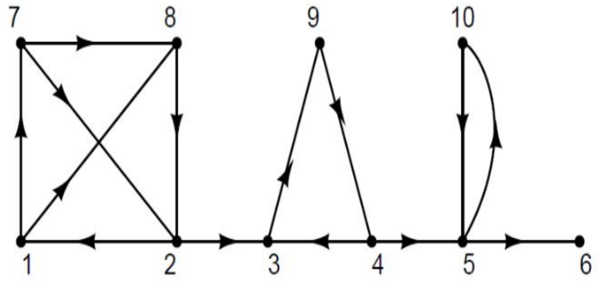
Week 6 Prims and Kruskal's Algorithm ()

Week 7 DFS and Cut Vertex ()

Week 8 Strong Connected Components & BFS ()

- Strong Connected Components (unit?unit=25&lesson=87)
- Strong Connected Components Part 2 (unit?unit=25&lesson=88)
- Strong Connected Components Part 3 (unit?unit=25&lesson=89)
- Strong Connected Components Part 4 (unit?unit=25&lesson=90)
- BFS (unit?unit=25&lesso

3) Let G be the ___ for the given graph.

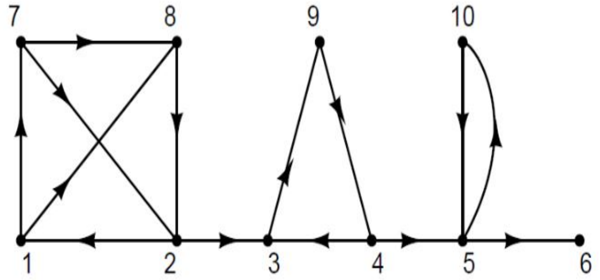


The number of edges in the condensed DAG is ___

3

2 points

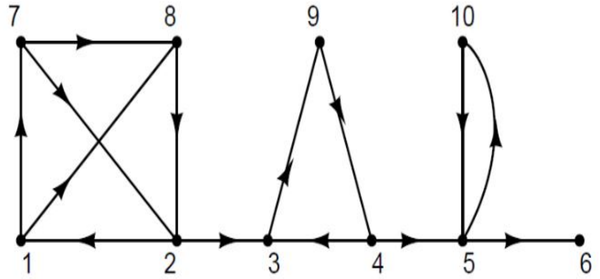
4) Suppose we do a BFS in the given graph starting from vertex 1, then the number of leve are (assuming root is at level 0) __.



4

2 points

5) Suppose we do a BFS in the given graph G starting from vertex 2, then the number of levels in the BFS tree is __.



4

2 points

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers